

Ansh Madaan

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Education

Thapar Institute of Engineering and Technology
B.Tech

2023 – Present
CGPA: 9.17

Govt. Boys Senior Secondary School
Class XII (PSEB)

2023
95.4%

Technical Skills

- **Languages:** Python, C/C++, R, SQL
- **Developer Tools:** Git, VS Code, Visual Studio
- **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib
- **Core Concepts:** Data Structures & Algorithms, OOP, Problem Solving

Projects

EEG-based Schizophrenia Detection using Deep Learning

[GitHub](#)

- Developed a hybrid **CNN–LSTM with attention** model achieving **97% accuracy** with subject-wise validation.
- Applied **bandpass filtering (1–50 Hz)** and temporal windowing for robust EEG feature extraction.
- Improved interpretability using **Grad-CAM, Integrated Gradients**, and attention weight visualization.

Face Recognition-Based Chatbot Authentication System

[GitHub](#)

- Built a secure chatbot access system using real-time face recognition with **OpenCV**.
- Integrated facial authentication with conversational AI for personalized interaction.
- Ensured high-accuracy face matching with efficient encoding and secure data handling.

Voice Assisted Terminal

[GitHub](#)

- Engineered a voice-controlled terminal enabling execution of complex shell operations via natural language.
- Integrated Google's generative AI models for intelligent command understanding.
- Published as a **PyPI package**, enabling easy installation and improved developer productivity.

Achievements

- Winner, **ACM Projectathon** — Led a team to victory among 100+ teams.
- Winner, **ACM CodeSprint** — Secured 1st place out of 50 teams.
- Recipient of **Merit Scholarship** for academic excellence (Top 5% at Thapar Institute).